

SAAD ZAIDI

AI & Software Engineer | BS Computer Science, FAST-NUCES

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PROFESSIONAL SUMMARY

AI/ML and Software Engineer with hands-on experience building production ML pipelines and scalable backend systems. Skilled in Python, PyTorch, scikit-learn, and Hugging Face Transformers for deep learning and anomaly detection, and in FastAPI, Flask, and PostgreSQL for RESTful API and full-stack SaaS development. Delivered end-to-end solutions spanning database design, LLM-powered automation, payment integration, and containerized deployment (Docker, Kubernetes). Dean's List student at FAST-NUCES with proven freelance and internship experience across ML engineering, backend engineering, and full-stack development.

CORE SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, Assembly

Machine Learning / AI: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Hugging Face Transformers, OpenAI API, Stability AI, SHAP, Streamlit

Backend Development: FastAPI, Flask, RESTful APIs, JWT Authentication, Celery, Redis, SQLAlchemy (async), Alembic, Pydantic

Frontend Development: React, TypeScript, Vite, Three.js, React-Three-Fiber, Valtio, HTML, CSS

Databases: PostgreSQL, MySQL, SQLite3, Supabase

DevOps & Tools: Docker, Kubernetes, GitHub Actions, Railway, Render, Vercel, Linux, Git

EXPERIENCE

AI/ML Intern, Saltec

Jun 2026 – Aug 2026

- Designed and built a rule + ML hybrid anomaly detection library covering 20+ use cases across RMS/sensor availability, energy-fault correlation (rectifier, DG, battery, breaker alarms), fuel theft and calibration drift, power theft, battery health/SOH, and solar underperformance for a multi-site telecom power infrastructure fleet.
- Engineered per-site dynamic thresholds, temporal gap handling, and windowed feature rollups (mean/min/max/slope/completeness) to separate genuine faults from sensor noise, data gaps, and normal operational variation across live telemetry streams.
- Built a weak-supervision pipeline that generates structured training labels from deterministic physical rules (fuel delta vs. refill/theft counters, DG runtime vs. grid/solar availability, battery depth-of-discharge vs. policy), removing the need for manual annotation at scale.

Freelance Software Developer — Backend & Full-Stack

2025 – 2026

- Delivered backend and full-stack web applications for multiple clients, independently owning architecture, deployment, and post-launch support (RO Plant Internal Portal, DesiBites POS, etc).
- 3DShopX (React, Three.js, React-Three-Fiber, Node.js, Express): built a real-time 3D AI-powered T-shirt customization platform enabling users to apply colors, textures, and decals to an interactive 3D model with multi-angle viewing; integrated Stability AI's text-to-image model to generate custom artwork applied directly to the shirt design; used Valtio for reactive state management and a Node/Express backend to handle AI image-generation requests.

PROJECTS

Anchor AI — ADHD Accountability Agent

2026

FastAPI, React, PostgreSQL, OpenAI, Celery, Redis, Paddle, Docker, Render, Vercel

- Architected and shipped a full-stack SaaS ADHD productivity agent delivering personalized task management via WhatsApp, including Google OAuth, phone OTP verification, Paddle subscription billing, and a dark neural-themed React dashboard.
- Designed an 18-table PostgreSQL schema with async SQLAlchemy, Alembic migrations, and Supabase hosting; implemented row-level cascade deletes, partial indexes, and JSONB columns for flexible preference storage.
- Built a GPT-4o Mini LLM pipeline for intent classification, task extraction, and micro-step generation; engineered Celery Beat scheduling for timezone-aware morning check-ins, task reminders, and EOD summaries across all users.
- Integrated Paddle Billing with HMAC-SHA256 webhook signature verification, idempotent event processing, and sandbox-to-production subscription lifecycle management (activated, paused, cancelled, resumed).

Intelligent CNC Process Monitoring & Defect Prediction

2026

PyTorch, XGBoost, scikit-learn, SHAP

- Developed a 3-layer ML pipeline: LSTM Autoencoder for anomaly detection ($F1 = 0.9316$, +43.3% over baseline), 1D-CNN + XGBoost for tool-wear classification (87.95% accuracy, $F1 = 87.86\%$), and stacked BiLSTM + Bahdanau Attention for RUL prediction ($R^2 = 0.5930$, +21.9%, MAE = 0.1343).
- Extracted 1,000+ FFT, statistical, and mechanical features from 18 GB of sensor data across three CNC datasets; applied SMOTE oversampling for class balancing and SHAP TreeExplainer for full feature-level explainability.

EDUCATION

FAST National University of Computer and Emerging Sciences — BS Computer Science

2023 – 2027

- Dean's List – 3rd & 6th Semester

CERTIFICATIONS

- LLM Engineering — Udemy (Ed Donner), 2024